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Environmental drivers of rhodolith beds and epiphytes community along the South Western Atlantic coast

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ABSTRACT

Environmental conditions shape the occurrence and abundance of habitat-building organisms at global scales. Rhodolith beds structure important hard substrate habitats for a large number of marine benthic organisms. These organisms can benefit local biodiversity levels, but also compete with rhodoliths for essential resources. Therefore, understanding the factors shaping the distribution of rhodoliths and their associated communities along entire distributional ranges is of much relevance for conservational biology, particularly in the scope of future environmental changes. Here we predict suitable habitat areas and identify the main environmental drivers of rhodoliths' variability and of associated epiphytes along a large-scale latitudinal gradient. Occurrence and abundance data were collected throughout the South-western Atlantic coast (SWA) and modelled against high resolution environmental predictors extracted from Bio-Oracle. The main drivers for rhodolith occurrence were light availability and temperature at the bottom of the ocean, while abundance was explained by nitrate, temperature and current velocity. Tropical regions showed the highest abundance of rhodoliths. No latitudinal pattern was detected in the variability of epiphytes abundance. However, significant differences were found between sampled sites regarding the composition of predominant taxa. The predictors influencing such differences were temperature and nitrate. The Tropical region is abundant in species with warm-water affinities, decreasing toward warm temperate region. The expressive occurrence of tropical species not referred before for warm temperate beds indicate a plausible tropicalization event.

1. Introduction

The global environmental patterns of temperature and irradiance drive the occurrence, demography and biodiversity levels of marine structuring species (Hillebrand, 2004; Spalding et al., 2007; Graham et al., 2007; Liuzzi et al., 2011; Freestone and Osman, 2011) such as mangroves (Rovai et al., 2016), seagrasses (Chefaoui et al., 2015), marine forests of macroalgae (Santelices and MARQUET, 1998; Keith et al., 2014; Graham et al., 2007; Assis et al., 2017; Bernardes et al., 2018) and rhodolith beds (Hernandez-Kantun et al., 2017). These organisms create complex three-dimensional structures providing essential habitats for a rich diversity and abundance of mobile (Ordines et al., 2015) and sessile fauna and flora (Steller et al., 2003, 2007; Sciberras et al., 2009; Peña et al., 2014).

Rhodoliths composed by non-geniculate red calcareous algae in free-living form (Foster, 2001; Pereira-Filho et al., 2011; Amado-Filho et al., 2012a), occurring from Tropical (Cavalcanti et al., 2014; Vale et al., 2018) to Polar Regions (Teichert, 2014), respond to temperature, nutrients and irradiance levels as a function of photosynthesis, calcification and respiration process (Schubert et al., 2019). Until a certain physiological threshold, these drivers improve photosynthesis and calcification in calcareous algae (Martin et al., 2013; Campbell et al., 2016; Comeau

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et al., 2016). For instance, a previous study indicated that under 30 °C and without subsaturating irradiance, rhodoliths could reach their highest metabolic rate (Vásquez-Elizondo and Enríquez, 2016; Hofmann et al., 2016). Thus, more suitable habitats for rhodolith growth are expected near tropical regions, where warmer waters with high transparency rates prevail. At local scales, moderate hydrodynamics allow rhodoliths to roll and keep a proper circular shape, while prevent the burial of the nodules caused by the excessive fine sedimentation (Foster, 2001; Sañé et al., 2016; Cabanellas-Reboredo et al., 2018). Water motion may further promote erosive processes and shape the associated community structure. Several studies correlated higher species richness and epibenthic biomass with reduced frequency and intensity of storms (Amado-Filho et al., 2007). Local nutrient load can also have an important role in determining the abundance of epibenthic species (Fong et al., 1994; Figueroa et al., 2014), with eutrophic regions showing substrates vastly colonized by fast growing macroalgae (Scherner et al., 2013; Neill and Nelson, 2016; Gouvêa et al., 2017). Some studies have further focused on the interactions of invertebrates and macroalgae with rhodolith beds (Scherner et al., 2010; Legrand et al., 2017). For instance, fleshy primary producers, as macroalgae can benefit local fauna by providing essential habitat and resources (Neill and Nelson, 2016; Aued et al., 2018). They can also protect rhodoliths from the excess of light in shallow waters (Figueiredo et al., 2000). However, in high quantity, macroalgae can shade the surface of host and compete for environmental resources and produce all ellopatical effects (Drake et al., 2003; Wahl, 2008).

Considering the crucial role of rhodoliths providing essential habitats across the global ocean, it is fundamental to study the drivers shaping the community structure of the beds and associated organisms, particularly in the scope of conservational biology in the faces of future environmental changes. The environmental conditions of South Atlantic coast are suitable for rhodolith beds (Foster, 2001). At the Brazilian coast, several beds have been found from 5°N (near Amazonian river; Moura et al., 2016), to 27°S (Santa Catarina state; Gherardi, 2004; Pascelli et al., 2013). At lower latitudes, rhodolith beds host a great biodiversity of macroalgae adapted to higher temperatures and irradiance (Riul et al., 2009; Bahia et al., 2010; Amado-Filho et al., 2012a; Foster et al., 2013; Amado-Filho et al., 2017), while at higher latitudes, species diversity decrease and replaced by cold adapted taxa (Pascelli et al., 2013). In warm temperate regions, diversity of rhodolith bed is also reduced (e.g., at Marine Protected Area of Arvoredo; Pascelli et al., 2013) suggesting that large-scale environmental drivers influence the abundance and occurrence of rhodoliths and associated macroalgae. However, the absence of standardized collected data provides weak frameworks to further discuss macroecological drivers.

The present study used depth standardized scuba dive sampling and macroecological modelling to explain and predict the occurrence and biomass of rhodolith beds and associated epiphytes along a large-scale latitudinal gradient in Southwestern Atlantic coast. High resolution environmental variables were extracted from Bio-Oracle 2.0 and used to model the variability of rhodoliths and macroalgal community. Our main hypotheses are that (1) even if the major part of the South Atlantic coast have environmental conditions that favor the rhodolith bed occurrence, the rhodolith abundance increases from higher to lower latitudes, where the high temperature and transparency of the water favor the photosynthesis and calcification of the tropical rhodolith-forming species and (2) the occurrence and abundance of epiphytes adapted to higher-transparency and warm-waters decrease from Tropical region to Warm-temperate region.

2. Material and methods

2.1. Study area

Sampling was conducted in the tropical, transition and warm temperate regions of the southwestern Atlantic Coast (according to Horta et al., 2001 and Spalding et al., 2007), between latitudes of ∼3°S and 27°S (sampling sites are depicted in Fig. 1; sampling coordinates and years in Table 1). The classification of Horta et al. (2001) in these three regions was based on phycological composition; the differences between regions were driven by habitat heterogeneity and temperature. Spalding et al. (2007) also drawn the same classification scheme, however considering additional marine organisms and forcing agents (e.g.: upwelling, nutrients, temperature, currents and bathymetry).

Starting from the surface layers up to the thermocline level, the most important current flowing along the Brazilian coast is the Brazil Current (BC) (Cirano et al., 2006). This current (BC) originates from the bifurcation of the Equatorial South Current (ESC), south of 10°S, and flows south, bordering the South American continent to the region of the Subtropical Convergence, located between 38°S ± 2° (Olson et al., 1988), where it forms the confluence with the Malvinas Current (MC) and moves away from the coast. The ESC also gives rise to the Northern Brazil Current (NBC) (Stramma, 1991; Silveira et al., 1994), which, in turn, flows towards the equator. The BC is associated to the movement of two masses of surface water (Tropical Water - TW and South Atlantic Central Water - SACW) (Cirano et al., 2006). There is also another water mass, formed by the runoff of continental waters (river water and estuarine plumes), called Coastal Water (CW, Castro et al., 2006).

According to Emilson (1961), TW is characterized by waters with temperatures higher than 20 °C, salinities above 36 ups and relatively low concentrations of dissolved nutrient, being transported by the BC off the northeast coast of the country to the north of Uruguay and Argentina. According to Miranda (1985), SACW is characterized by temperatures above 6 °C and below 20 °C, salinities between 34.6 and 36 ups and rich in nutrients. SACW is formed by the meeting of currents that form the Subtropical Convergence and transported to the south by the Brazil Current, often resurfaces at some points off the coast near the surface through upwelling phenomena (Matano et al., 2010).

Rata Island (FN), Itaparica (ITA) and Morro de São Paulo (MSP) are located in the tropical region. This is influenced by TW, which is transported by BC and NBC (Stramma and England, 1999; Castro et al., 2006) and CW. The major part of the coastal shelf is relatively narrow, varying from 8 km offshore of Salvador to 200 km in Abrolhos (Knoppers et al., 1999). The sea surface temperature (SST) in this region varies from 25 to 29 °C.

Guarapari (GUA) and Itaipava (SE) are located in the transition between the Tropical and Warm Temperate regions. This region is influenced by CW and also by BC (Castro and Miranda, 1998). Despite the predominance of oligotrophic waters in this region, with SSTs varying from 21 to 27 °C, a seasonal upwelling of the SACW reaches the southern margin during summer (Schmid et al., 1995). The high abundance of hard substrates and the seasonal cold-water upwelling with high concentration of nutrients favors the richest marine flora of Brazil, with occurrence of common species from the two adjacent regions (Guimarães, 2003; Amado-Filho et al., 2007).

Deserta (DE) and Rancho Norte (RN) are located in the Warm Temperate region within the Arvoredo Biological Marine Reserve (Rebio Arvoredo). These rhodolith beds represent the southernmost limit of distribution of rhodoliths in the Western Atlantic (Gherardi, 2004; Pascelli et al., 2013), where there is complex oceanographic dynamics due to strong winds and Subtropical Convergence (Freire et al., 2017). The confluence of warm and tropical waters (TW) transported by BC with cold waters transported by the Malvinas Current (MC) from the south, give rise to SACW (Peterson and Stramma, 1991). Frequently, the SACW resurfaces on the coast near the surface during the summer by lowering the temperature, salinity and exporting many nutrients to the surface waters. In addition, in the nearby of Rebio Arvoredo, the local freshwater outflow from the Tijucas river or the brackish waters of the Santa Catarina Island Channel also alter the characteristics of seawater (Freire et al., 2017). Eventually, the waters originating from the La Plata river also reach the region (Möller et al., 2008; Strub et al., 2015). All this oceanographic dynamic influence the thermal amplitude of the SST

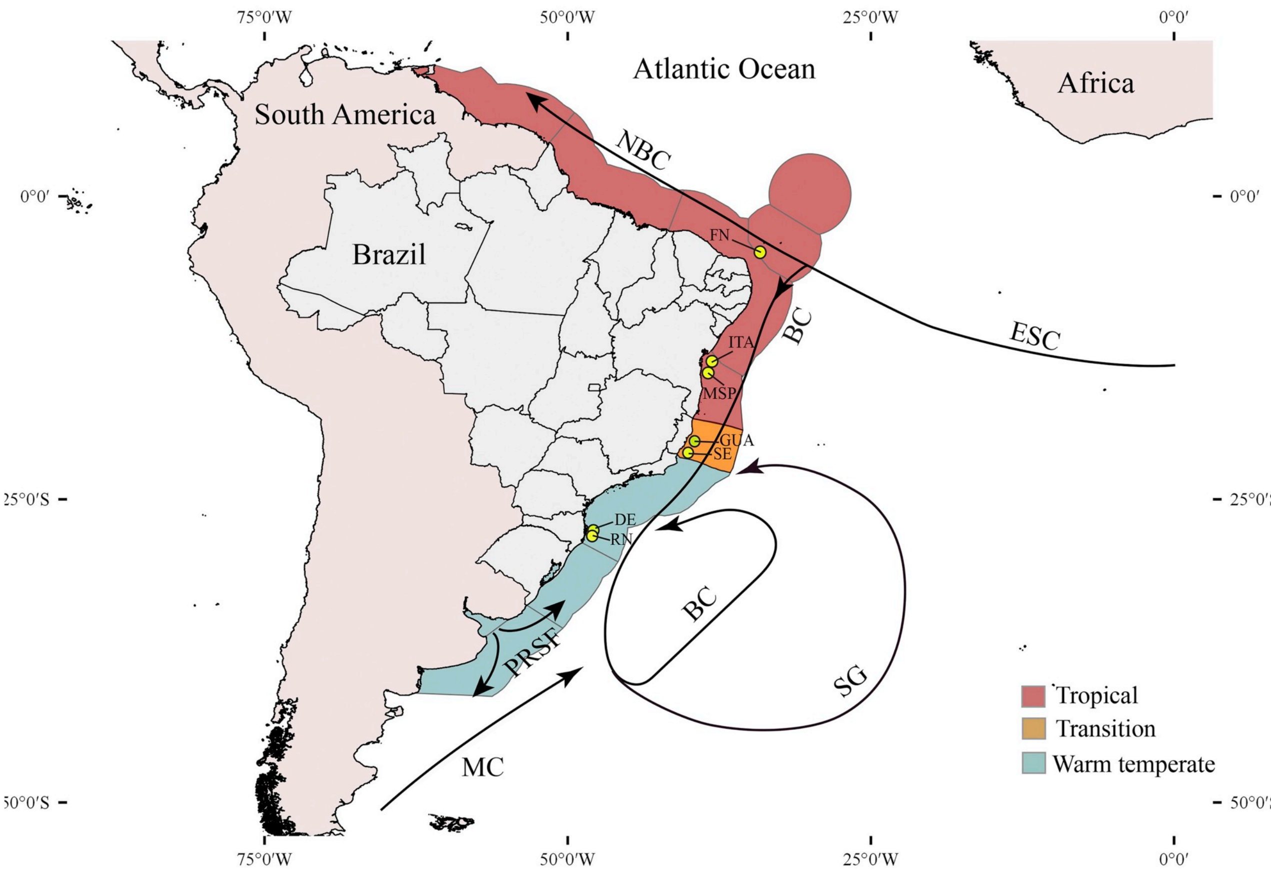

Fig. 1. Sampling points (code) along the tropical, transition and warm temperate regions of the southwestern Atlantic (FN: Rata island; ITA: Itaparica; MSP: Morro de São Paulo; GUA: Guarapari; SE: Sequim; DE: Deserta Island; RN: Rancho Norte). The arrows indicate Northern Brazil Current (NBC), Equatorial South Current (ESC), Brazil Current (BC), Subtropical Gyre (SG), Prata River Sediment Fluxes (PRSF) and Malvinas Currents (MC). Adapted from Horta et al. (2001) and Spalding et al. (2007).

Table 1
Sampling points along the southwestern Atlantic with information for site name, coordinates, year of sampling and number of quadrats (NB).

City (state)/Site (code)	Coordinate	Year of sampling	NB
Fernando de Noronha (PE)/Rata Island (FN)	03°53′S; 32°28′W	Summer/2016	5
Itaparica (BA)/Itaparica Island (ITA)	13°04′S; 38°40′W	Summer/2014	15
Cairu (BA)/Morro de São Paulo (MSP)	13°25′S; 38°52′W	Summer/2014	15
Guarapari (ES)/Rasas Island (GUA)	20°40′S; 40°21′W	Summer/2014	15
Itaipava (ES)/Sequim (SE)	20°59′S; 40°44′W	Summer/2014	14
Florianópolis (SC)/Deserta Island (DE)	27°15′S; 48°19′W	Summer/2014	13
Florianópolis (SC)/Rancho Norte (RN)	27°16′S; 48°22′W	Summer/2015	11

that varies from 16 to 27 °C and it is also responsible for the increase of the primary production and fishing in the region (Freire et al., 2017).

2.2. Sampling and processing

The sampling campaigns were conducted in rhodolith beds during summer and at a standardized depth of 10 ± 1 m. These depth and season was chosen due to higher abundance of rhodolith and associated macroalgae in shallow waters (Riul et al., 2009; Amado-Filho et al., 2007; Pascelli et al., 2013). Rhodoliths and associated epiphytes were collected within 15 quadrats of 0.0625 m^2, 2 m distant from each other, randomly disposed along the beds. Due to adverse weather conditions during sampling this number was not reached at some sites (detailed at Table 1). The contents of each quadrat were taken to the laboratory and preserved at −20 °C until triage and identification at the species level of epiphytes following standard methods (Joly, 1967; Cordeiro-Marino, 1978; Littler and Littler, 2000) and updates considering most recent information about taxononomy and nomenclature regarding each group (e.g. Oliveira-Carvalho et al., 2010; Nauer et al., 2014; Jamas et al., 2017). To estimate the mass of rhodoliths, all of the rodoliths of each quadrat were dried at 60 °C for 48 h and weighted (balance precision of 1×10^{-3} g). The same procedure was adopted to measure the biomass of separated species of epiphytes. The genera *Neosiphonia, Polysiphonia, Ceramium* and *Centroceras* frequently occurred together on turf assemblage. Because of this, they were weight together and considered as Filamentous Red Algae in the analyses. The rhodoliths were disposed in a beaker filled with water to estimate volume by water displacement (Pascelli et al., 2013). The relation between dry mass (g DW) and volume (cm^3) of rhodoliths was used to determine density (g DW. cm^{-3}) within each quadrat.

2.3. Statistical analyses

The differences between volume, density and dry mass of rhodoliths and epiphytes between sampling sites were analyzed by Kruskal-Wallis One Way Analysis of Variance with Dunn multiple comparisons of mean ranks, using Bonferroni correction (for multiple comparisons).

Models explaining and predicting the potential occurrence and dry mass of rhodolith beds were developed using a delta-lognormal statistical approach as presences or absence). This fitted environmental predictors against the occurrence of rhodoliths (i.e., binomial response variable), separately from dry mass (i.e., lognormal variable), and combined both responses with a product function (reviewed by McGill et al., 2007). Such integrative approach allows predicting dry mass only in regions where presences were inferred by the binomial model. To this end, the machine-learning algorithm Boosted Regression Trees (BRT) was used since it handles non-linear relationships and complex interactions, while avoiding overfitting (i.e., the fit describing random noise) by forcing predictors to have positive or negative monotonic responses on the models (i.e., constraining the modelling fit to have an increasing or decreasing relationship), and by optimizing the number of trees, tree complexity and learning rate (for more details refer to Elith et al., 2008). A 4-fold cross-validation framework using independent latitudinal bands was implemented to tune the three main parameters (e.g., Neiva et al., 2015; Assis et al., 2016, 2017).

The environmental predictors used in the analyses were selected considering the ecological knowledge of coralline algae (e.g., Martin et al., 2014; Sañé et al., 2016), and were extracted from Bio-Oracle (Tyberghein et al., 2012; Assis et al., 2018), an open source dataset with information for the bottom of the ocean at a spatial resolution of 30 arcmin (~9.2 km at the equator). Considering the high correlation found between nitrates and phosphates (0.94; Fig. S1), we discarded the latter from analyses since it showed an higher correlation with maximum temperatures (0.97; Fig. S1).

The binomial model used georeferenced occurrence data gathered from the Global Biodiversity Information Facility (GBIF) (http://data.gbif.org) and the available literature (Table S1), alongside with pseudo-absences generated with a three-step technique for improved species distribution modelling (see Senay et al., 2013 for details). The lognormal model used the mean dry mass of each site, considering all samples available.

True skill statistics (TSS) and the area of the receiver operating characteristic curve (i.e., AUC) were used to assess the performance of the binomial model while deviance explained was used for the lognormal model. The ecological significance of both models was investigated by determining the contribution of each environmental predictor to the overall performance of models and by developing partial dependency plots for each predictor. These were made by fixing all alternative predictors onto their average (Elith et al., 2008).

The differences between the three regions with respect to community composition of epiphytes were analyzed using PERMANOVA, based on Bray-Curtis dissimilarities using 999 permutations (Anderson, 2001). The BIO-ENV procedure (Clarke and Ainsworth, 1993) used to identify which combination of the environmental variables (temperature, nitrate, current velocity and light) extracted from Bio-Oracle best explained variations in community structure of epiphytes (Clarke and Ainsworth, 1993). SIMPER analysis was conducted to identify the species that most contributed to Bray–Curtis dissimilarity between regions (Clarke, 1993). The results of the BIO-ENV procedure were represented through in Non-metrical multidimensional scale (nMDS) ordinations of the states based on each composition of epiphytes (Bray–Curtis dissimilarities) and the associated best subset of explanatory environmental variables (Euclidean dissimilarities). Non-parametric Spearman test were also used to determine correlations between environmental predictors. These statistical analyses were performed using R software and packages "vegan".

3. Results

3.1. Rhodoliths abundance

The dry mass, volume and density of rhodoliths from each sampling site were significantly different ($H = 56.80461$, $p < 0,001$; $H = 32.43684$, $p < 0.001$; $H = 54.07191$, $p < 0.001$, respectively). The highest dry mass and volume corresponded to the tropical site FN (36,476.3 [±10,295.9] gDW m^{-2} and 26,080 [±8077.2] cm^3 m^{-2}, respectively) and the lowest to warm temperate DE (11,121.4 [±2930.7] gDW m^{-2} and 11,729.2 [±3936.0] cm^3 m^{-2}) and transition site SE (9652.8 [±3558.0] gDW m^{-2} and 10,858.6 [±5475.1] cm^3 m^{-2}, Fig. 2A and B). Regarding density (Fig. 2C), the results were more homogeneous. The highest density corresponded to tropical sites MSP (1.44[±0.22]), ITA (1.39[±0.06]) and FN (1.40[±0.09] gDW cm^{-3}).

3.2. Environmental drivers of rhodolith beds

The ecological models developed to explain and predict the occurrence of rhodoliths retrieved high performance (TSS: 0.90, AUC 0.95). In the same way, the model for dry mass showed little deviation (deviance explained: 0.90) between observations and predictions (average difference of 2519 ± 2086 gDW m^{-2}; Fig. 5; Fig. S2). The contribution of predictors to the models showed the occurrence of rhodolith beds (at the scales of our study) being mainly determined by maximum and minimum temperatures, as well as the light availability at the bottom of the ocean (contributions > 10%; Fig. 3). The partial dependency plots showed suitable habitats between ~16.0 and ~28.8 °C, although with higher probability between ~17.5 °C and ~26.5 °C, and light availability above ~3E. m^{-2}. yr^{-1} (Fig. 4). Where rhodoliths occur, dry mass was largely explained by nitrate availability, and on a lower degree, by current velocity and minimum ocean temperatures (Fig. 3). Higher levels of dry mass were explained by temperatures above 23.3 °C, nitrates above 2 mmol m^{-3} and current velocity above 0.22 m s^{-1}. The integration of both models (i.e., delta-lognormal approach) predicted 229,718 km^2 of suitable habitats distributed between Rio Grande do Sul and Amapá States (Brazil) and between 1 and 149m (Fig. S3), with a total potential overall dry mass of 4560×10^6 ton.

3.3. Epiphytes abundance

The total epiphytes biomass was significantly different among sites ($H = 45.54$, $p < 0.001$), however with no clear latitudinal pattern (Fig. 2D). ITA had the highest biomass, with 17 (±22) gDW m^{-2} and GUA the lowest, with 0.412 (±0.240) gDW m^{-2}. With respect to the composition of epiphytes species, there were significant differences ($F = 8.77$, $p < 0.001$) between tropical, warm temperate and transition regions (Fig. 6). SIMPER analyses showed the macroalgae *Dictyopteris jolyana* mostly responsible for the differences between tropical and warm temperate regions (contribution: 31.17%), while *Caulerpa pusilla* contributed to the differences between tropical and transition regions (contribution: 12.58%) and between warm temperate and transition regions (contribution: 17.10%; Table S4). The BEST analysis of BIO-ENV procedure indicated that the setting of epiphytes that most contributed to the differences in the overall community was composed by Filamentous Red Algae group, *Hypnea spinella*, *Jania* spp, *Dictyota bartayresiana*, *Canistrocarpus cervicornis*, *Dictyopteris jolyana*, *Sargassum* sp. and *Padina gymnospora* (correlation = 0.67; Fig. 6). The environmental predictors that explain such variation in community structure were maximum temperature and minimum nitrate (correlation = 0.26).

4. Discussion

The revealed community structure, abundance of rhodoliths and the suitable area around 230,000 km^2 provide a new magnitude to Brazilian rhodolith bed importance as major biofactory of carbonate of the earth (Amado Filho et al., 2012b). This ubiquitous suitable area corroborates the deposit of 2.10^{-11} tons of carbonate off the Brazilian coast (Milliman and Amaral, 1974). Despite, studies focusing on the composition and structure of rhodolith beds have become increasingly available during the last years, this is the first to focus on its abundance and associated community structure along a broad latitudinal gradient. By combining standardized sampling with macroecological modelling, it shows a clear latitudinal pattern of rhodolith abundance mainly explained by

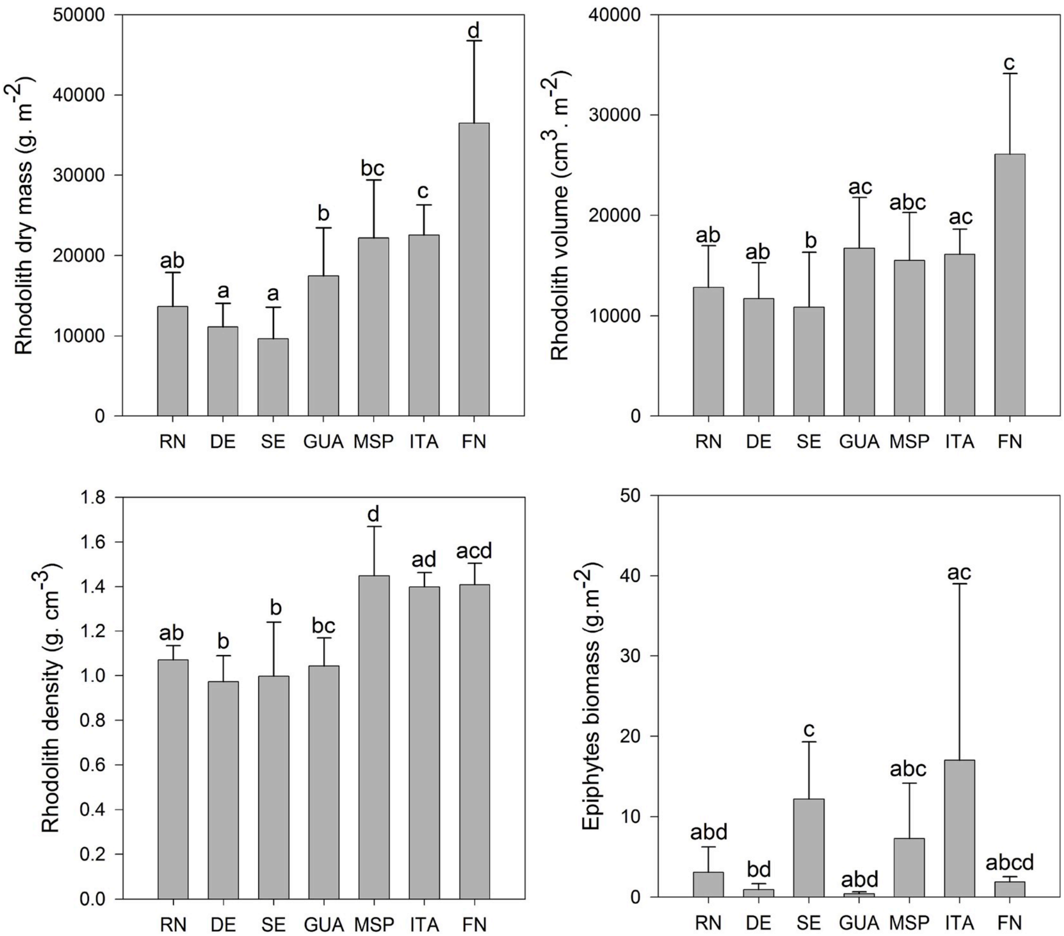

Fig. 2. Dry mass (A), volume (B) and density (C) of rhodoliths and biomass of epiphytes (D) from seven collections points along Brazilian coast (±SD bars). RN: Rancho Norte (Florianópolis, SC); DE: Deserta Island (Florianópolis – SC); SE: Sequim (Itaipava – ES); GUA: Rasas Island (Guarapari – ES); MSP: Morro de São Paulo (Cairu– BA); ITA:Itaparica Island (Itaparica– BA); FN:Rata Island (Fernando de Noronha– PE). Letters above bars indicate results of Kruskall-Wallis multiple comparisons test ($p < 0.05$).

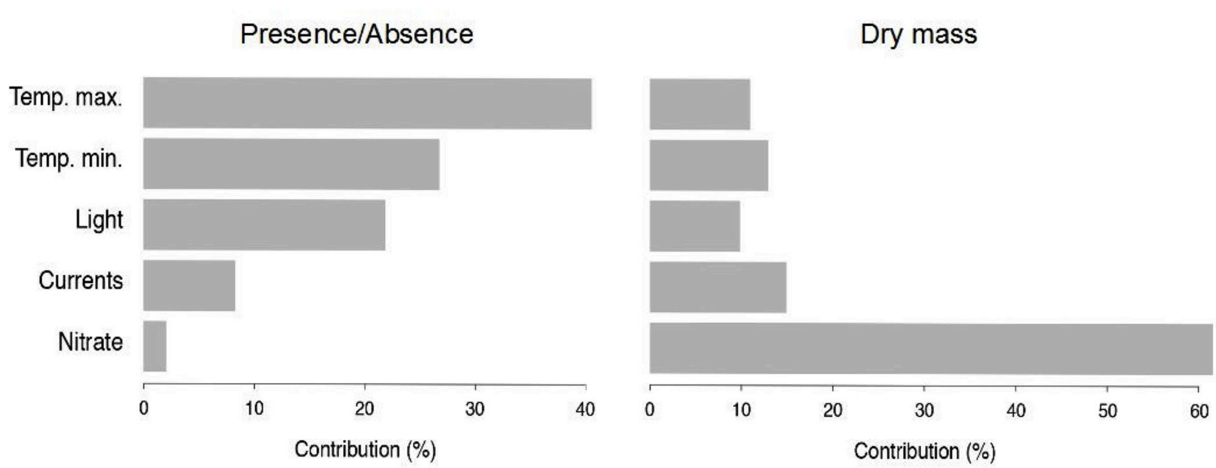

Fig. 3. Relative contribution of climate predictors to the models developed for (left panel) presence/absence and (right panel) dry mass of rhodoliths.

nutrients, current velocity, temperatures, and light. With respect to total biomass of associated macroalgae, there was no evidence for a latitudinal gradient; however, the differences found in species composition corroborate previous biogeographical subdivisions (Horta et al., 2001), which are explained by the higher temperatures and irradiance levels of tropics or the higher nutrient load in the warm temperate region. According to the results, while the abundance of rhodoliths increased with temperature, the limit of 28 °C is yet preferable for occurrence, a threshold corroborated by previous ecophysiological experiments performed on tropical species (Diaz-Pulido et al. 2012; Vásquez-Elizondo and Enríquez, 2016; Scherner et al., 2016). Considering the extension and biological importance of these formations along the Brazilian coast,

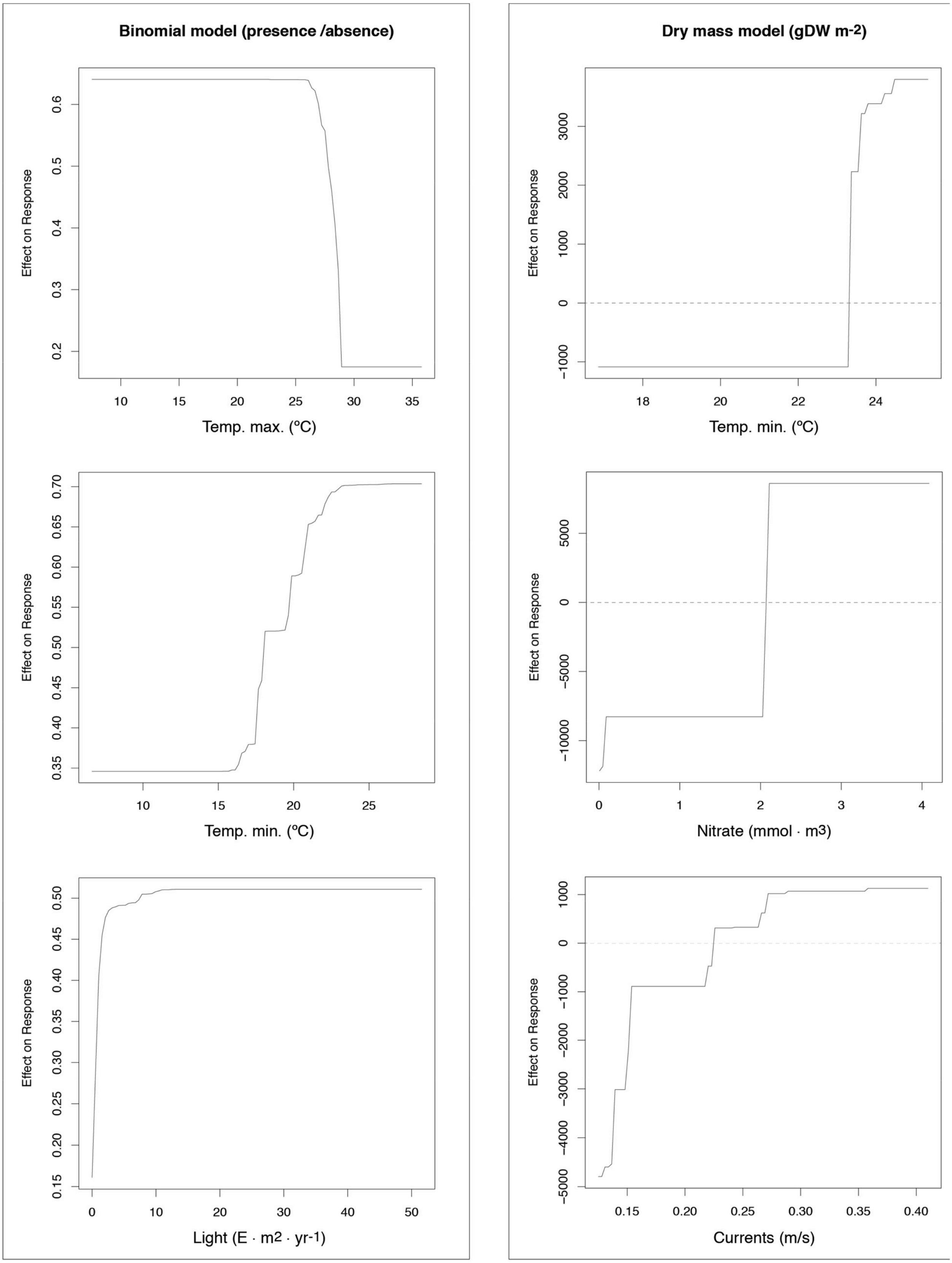

Fig. 4. Partial dependency functions depicting the effect of the most important predictors on the models. The effect of temperatures (maximum and minimum) and light on the binomial response predicting presence or absence of rhodoliths (left panel) and the effect of minimum temperature, nutrients (as nitrate) and current velocity on the predicted dry mass of rhodoliths (right panel).

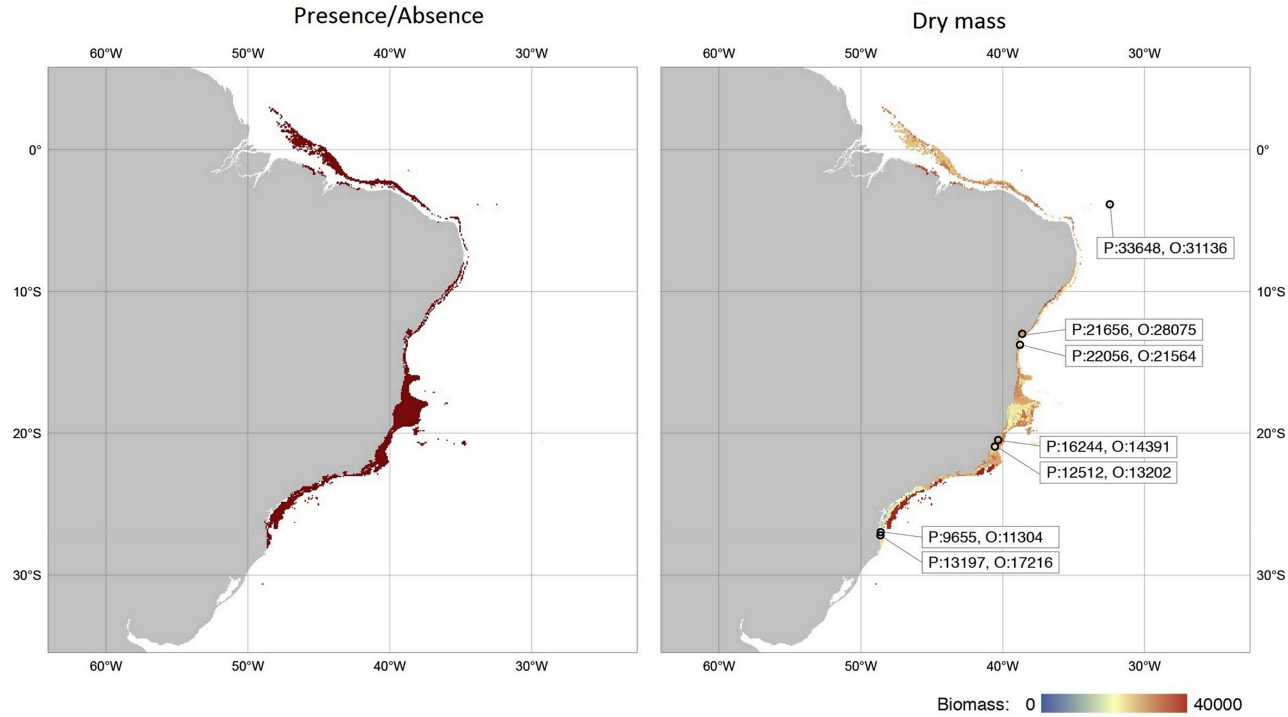

Fig. 5. Potential distribution (left panel) and dry mass (right panel) of rhodolith beds (g/m^2). Black circles (right panel) depict sampling sites and boxes the correspondent predicted (P) and observed (O) dry mass ($gDW\ m^{-2}$).

comparable to the global coral reef (Jones et al., 2015), reinforces the need for further discussion regarding their conservation and management regarding global and local threats.

4.1. Rhodolith abundance

The variation of abundance and density of rhodoliths followed the explored latitudinal gradient, with higher values in tropical sites decreasing towards warm temperate. The higher dry mass in the tropics indicate a larger potential of this region for carbonate production and habitat building (Steller et al., 2003; Amado-Filho et al., 2012b; Foster et al., 2013), while the higher density suggests more robust and heavy individuals than in the warm temperate region. These characteristics may protect them from potential breaks caused by strong water movement, favoring the stability of the bed (Marrack, 1999; Hinojosa-Arango and Riosmena-Rodríguez, 2004; Amado-Filho et al., 2007). Some physical aspects of the environment are known to influence the morphology and internal structure of rhodoliths (Basso, 1998), which can lead to different density. Depending on the frequency of turning caused by organisms or local currents, rhodoliths can assume different forms, such as densely branched and laminar when subjected to a frequent turning and high energy conditions, or columnar when subjected to stabilized conditions (Bosence, 1983a., 1983b). Morphological analyses integrated with internal structure reveal that the frequency of overturning of rhodoliths increases from the 'boxwork' (with internal macroscopical voids) to maerl facies and pralines (without macroscopic cavities) (Basso, 1998). These aspects can suggest that the presence or absence of internal cavities of rhodoliths can be the cause of the differences in densities.

The ecological models indicated that their occurrence is driven by increasing temperatures (until 28 °C) and by light availability, while nutrients are the main predictor explaining abundance. While the effect of temperature on calcification and photosynthesis of coralline algae are not yet well known at global scales, they have been reported for tropical (Comeau et al., 2016), temperate (Martin et al., 2013) and polar environments (King and Schramm, 1982). For instance, the calcification in *Lithophyllum kotschyanum* increased linearly with temperature during spring, except when exposed to a thermal maximum of 31.5 °C (Comeau et al., 2016), a threshold close to that inferred in our study. Moreover, the gross photosynthesis in the Mediterranean crustose coralline algae *Lithophyllum cabiochiae* was about two to threefold higher in summer than in winter, while calcification was four to eightfold higher in summer than in winter (Martin et al., 2013). In the Baltic Sea, the calcification of *Phymatolithon calcareum* increased linearly from 10 to 24 μg $CaCO_3\ g^{-1}\ h^{-1}$ between 0 and 20 °C (King and Schramm, 1982). The evaluation of *L. crispatum* and *M. erubescens*, two southern Atlantic widespread species revealed that C assimilation during photosynthesis is temperature and nutrient dependent, while the calcification change in accordance with the species and temperature. These results reinforce the model output and standout these two drivers as key environmental factors limiting rhodolith distribution and abundance (Schubert et al., 2019). The positive physiological response of coralline algae to temperature increase (until a certain limit) indicates that warm tropical sites are more suitable to rhodolith bed growth. However, the populations from their respective biogeographical provinces are exposed to different selective pressures. As described by Schubert et al. (2019), warm temperate populations, that present some minor molecular differences from tropical relatives (Sissini et al., 2014), when exposed to temperatures observed in tropical provinces reduced their photosynthesis and calcification.

Therefore increase our knowledge about rhodolith taxonomic particularities should enhance our understanding of regional variability and functioning. An inventory made by Amado-Filho et al. (2017) indicated 33 rhodolith-forming species in Brazil, compared with 12 in Europe and Azores, 7 in Australia and New Zeland, 4 in Gulf of California, 1 in Alaska and 1 in Canada. Beyond the temperature, the light availability provided by the transparency of tropical waters seems to favor the occurrence and biomass of rhodolith beds. At Mediterranean Sea, increasing irradiance until a certain limit improves photosynthetic rates, which consequently enhanced calcification (Martin et al., 2013).

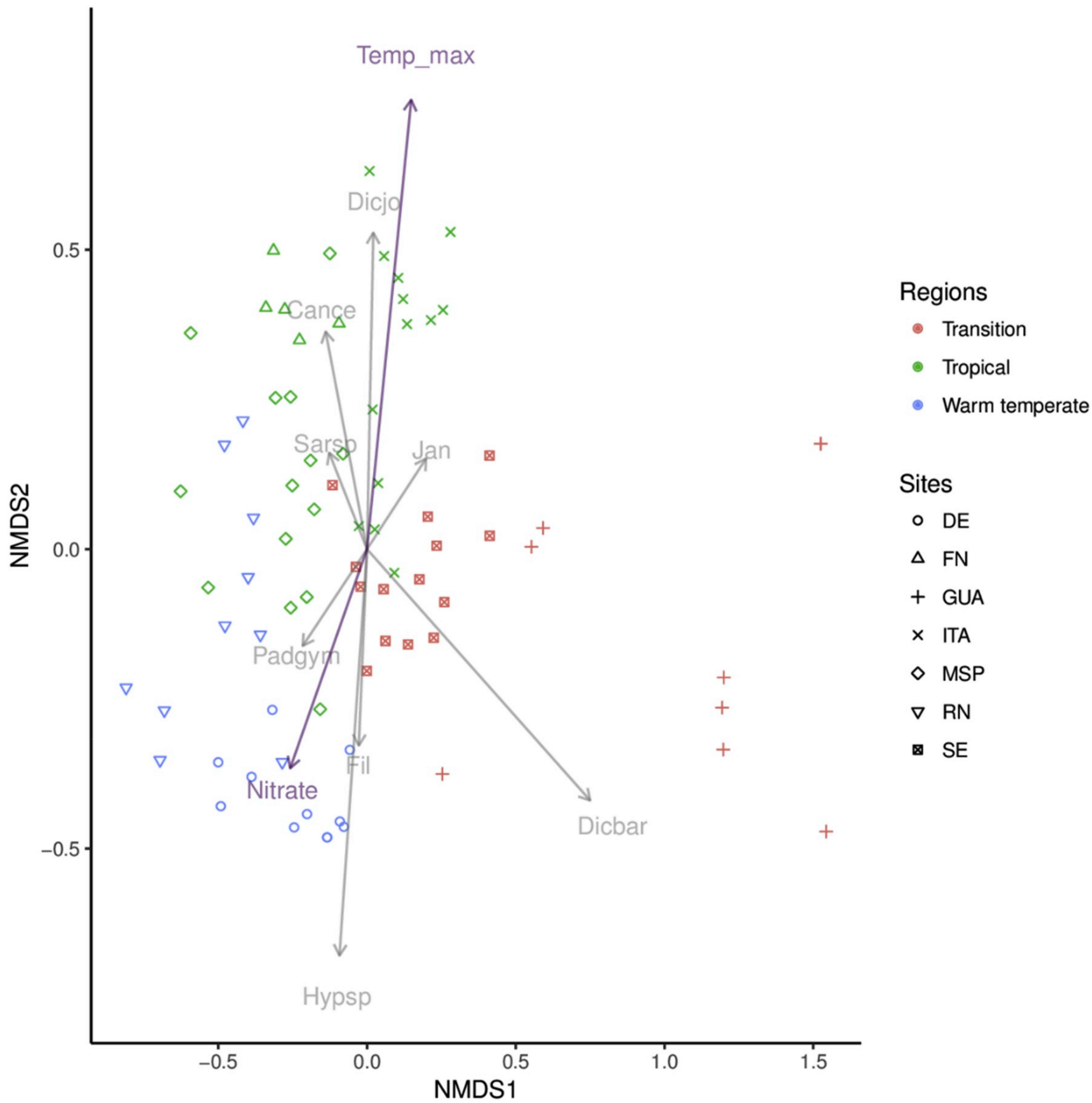

Fig. 6. nMDS (stress = 0.161) representing groups of epiphytes community of rhodolith beds from Sites at Brazilian Regions highlighted in different colors ($p < 0.001$). Gray arrows indicate species that most contributed to differences among communities (Codes: Padgym = *Padina gymnospora*, Dicjo = *Dictyopteris jolyana*, Jan = *Jania* sp., Dicbar = *Dictyota bartayresiana*, Hypsp = *Hypnea spinella* and Fil = Filamentous red algae, Sarsp = *Sargassum* sp., Cance = *Canistrocarpus cervicornis*). Purple arrows indicate environmental variables correlated with each species composition. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Crustose coralline algae can exhibit a metabolic control of surface pH through photosynthesis and light-inducible H^+ pumps, which play an important role in biological processes related to inorganic carbon uptake and calcification (Hofmann et al., 2016). In Polar Regions, moderate light is the important factor to $CaCO_3$ increment, followed by temperature (Teichert and Freiwald, 2014). In *Lithothamnion glaciale*, the summer growth seems to depend on the effect of previous winter cloud cover (Burdett et al., 2011). However, to deal with excessive light levels that can cause damage in photosynthetic apparatus, some few red coralline algae that colonize tropical regions (i.e. *Porolithon* sp. and *Lithophyllum kotschyanum*) can adopt dynamic photoinhibition strategies to tolerate and minimize the photodamage, and optimise photosynthesis (Burdett et al., 2014).

Regarding nutrients, the positive correlation between nitrates and dry mass agrees with Dutertre et al. (2015) that also reported an increase in percentage of maerl cover near areas with higher nitrate concentration. In the same way, additional experiments showed the growth of the coralline algae *Lithophyllum yessoense* positively correlated with nitrate concentrations (Ichiki et al., 2000). However, Bjork et al. (1995) reported no significant effect of nitrate or ammonia on coralline algae calcification, while Vale et al. (2018) reported a reduction of rhodoliths near areas with high organic matter loads near Amazon River. However, our model showed no probability of occurrence in that area related to reduce light availability due to very high of sedimentation, in addition to reduction of salinity by important inflow of freshwater, and not due to increasing nutrients. This may also explain the lower observed and predicted dry mass adjacent to the coast of Santa Catarina (at higher latitudes), which receive the plume coming from Rio de La Plata, Central Water of South Atlantic and, eventually, from the Malvinas Current, alongside with other local sources of continental runoff (Möller et al., 2008; Strub et al., 2015). These coastal water also present lower pH (Kerr et al., 2016), what reduce calcification and carbonate content (Rich et al., 2018). In this region, higher dry mass of rhodoliths were predicted 60 km offshore, likely due to irradiance availability on deeper areas and favorable temperature and nutrient levels.

Currents were also important to explain the dry mass of rhodoliths, although on a lower extent. Regions exposed to intense currents provide the necessary movement to avoid epiphytes and fauna overgrowth, which cover the rhodolith surface and disturb the acquisition of resources. Martin et al. (2014) suggests that flat and coarse-grained areas with strong bottom currents tends to be suitable habitats to rhodolith beds in the Mediterranean Sea, since that reduces sedimentation. Moderate hydrodynamics are preferable to rhodoliths than lower or higher (Cabanellas-Reboredo et al., 2018; Melbourne et al., 2018). Low hydrodynamics favors sedimentation, disrupts gas exchange, and enables overgrowth of epiphytic organisms under rhodolith surface (Steller and Foster, 1995; Sañé et al., 2016). Similarly, a more active hydrodynamic regime may also increase rhodolith density. The greater solidity of *Spongites fruticulosus* rhodoliths has been attributed to wave action that polishes the edges of this nucleated species (Steller and Foster, 1995; Cabanellas-Reboredo et al., 2018).

4.2. Epiphytes abundance

The differences in epiphytes community composition were due to the predominance of species with tropical and warm temperate preferences. The relevant species structuring the community of the tropical region (driven by higher temperatures) were *D. jolyana* and *C. cervicornis.* Dictyotales were previously referred as abundant in the tropics (Sangil et al., 2011; Brasileiro et al., 2016). They can grow in places with higher light levels since they have biochemical mechanisms to dissipate the energy surplus and mitigate photodamage (Celis-Plá et al., 2015). At warm temperate sites, the abundance of tropical genera decreases and tends to be replaced by species that have cold-water affinities. A latitudinal variation in seaweed communities was also reported for NE Atlantic, with warm-tolerant species in south and cold and low-light-tolerant species in north (Peña et al., 2014). As shown in results, *Hypnea spinella* and Filamentous red algae were relevant for the community of this region, positively correlated with minimum nitrate. Studies with *Hypnea* species, indicate that these algae can tolerate high nutrient concentration and act as biofilters of nitrogen and phosphorus (Martins et al., 2009; Ribeiro et al., 2013, Whitehouse and Lapointe, 2015). Moreover, phosphate, which was correlated with nitrate (Fig. S1), has a role in metabolic processes related to nitrate reductase activity, responsible to catalyze the reduction of nitrate to nitrite, which is an important factor in regulation of growth and protein production (Martins et al., 2009). In general, previous studies observed that algae with higher surface-area/volume ratio tend to have more nutrient requirement and faster growth than thicker algae (Taylor et al., 1998; Piazzi et al., 2011).

Due to the peculiarities of ES state, it was referred as a transition zone between Tropical and Warm temperate regions of Brazil with respect to macroalgae flora (Horta et al., 2001; Guimaraes and Amado-Filho, 2008; Amado-Filho et al., 2010) and other marine organisms (Teixeira et al., 2013; Vila-Nova et al., 2014). Our work reinforces this qualitative pattern. The species that most contributed to the differences found in this region was *C. pusilla*. Additional studies with other species of *Caulerpa* revealed that they can survive in a wide range of temperatures (10–30°C, Terrados and Ros, 1992 or 15–30°C, Ukabi et al., 2013) and persist under high nutrient load (Lapointe and Bedford, 2010), which are the conditions reported for this region in summer. The transition zone, influenced by tropical waters from the north and seasonal cold and nutrient-rich waters from south, enables the co-occurrence of organisms tolerant to both tropical and warm temperate regimes (Guimarães, 2003).

Besides the predominance of the tropical *P. gymnospora* in the warm temperate site (RN), expressive biomass of other Dictyotales algae such as *Canistrocarpus cervicornis* and *Dictyota delicatula* reinforce some tropical affinities (Table S2). These species were not referred for RN rhodolith bed by Pascelli et al. (2013), who highlighted the low diversity of macroalgae of this region, relating this to warm temperate climate of the region. Therefore, these plausible shifts in flora composition suggest a response to recent environmental changes, making this region more similar to tropics. The tropicalization process due to climate change have been described for subtropical and temperate regions (Vergés et al., 2014; Wernberg et al., 2016; Araújo et al., 2018a). In fact, the sampling at RN during the summer of 2015 coincided with the occurrence of El Niño event, reflecting substantial environmental changes such as altered runoffs, winds and marine currents, as well as temperatures (Freire et al., 2017). Moreover, Gouvêa et al. (2017) registered a strong heatwave at warm temperate coast in the Spring of 2014, which could cause the shifts in community structure of our sampling. These events have been more frequent and intense due to climate changes (Wernberg et al., 2013, 2016).

Although several publications cited in this study highlight the importance of different abiotic characteristics of tropical and warm temperate environments to macroalgae growth, the correlation with the best environmental variable predictors was weak (0.26). This can indicate that faster and local-scale environmental changes not considered here could drive the abundance of these fast growth organisms (Riul et al., 2009; Pascelli et al., 2013; McConnico et al., 2017). It is important to highlight that the revealed pattern is limited to a summer evaluation. However, we recognize that part of the observed differences during this season is influenced by the selection of assemblages imposed by the limiting light, temperature and extreme storms imposed by the winter condition as referred by Pascelli et al. (2013). Therefore, environmental changes along seasons and years should add additional light in the discussion about environmental drivers of macroalgal community associated with rhodolith beds (Gatti et al., 2015).

Besides the abiotic factors considered, herbivory is a major driver of macroalgal composition (Littler et al., 1995; Poore et al., 2012; Bonaldo and Hay, 2014; Longo et al., 2015). The action of herbivores along latitudinal gradient seems to vary according to the organisms considered. The herbivory by fishes tends to increase towards the tropics, due to higher temperatures (Floeter et al., 2004; Ferreira et al., 2004; Longo et al., 2014). The abundance of invertebrates (which include herbivores) increases at latitudes higher than 20° in the southeastern Atlantic (Aued et al., 2018). Carvalho et al. (2018, unpublished data) observed the disappearance of epiphytes at RN rhodolith bed along time, following the increase in invertebrates. However, Poore et al. (2012) analyzed experiments with fishes and invertebrate herbivores exclusion and concluded that there was no latitudinal gradient of herbivory, being only dependent on the taxonomic (with major impact on brown algae) or morphological group. The dominant species found in tropical sites are Dictyotales members that can produce chemical defenses when subjected to herbivory (Stachowicz and Hay, 1996; Bianco et al., 2010; Araújo et al., 2018b), which suggest that herbivores could have an important role in structuring macroalgal community in this region.

Finally, the positive influence of higher temperatures on tropical Dictyotales abundance shows that these faster growth species could benefit from a possible temperature increment, and so does rhodoliths abundance. Although this study focused only on the abundance of these algae on rhodolith beds, they were reported for tropical shallower environments experiencing temperatures above 28 °C (Nunes and Paula, 2006; Scherner et al., 2013). Still, the occurrence of the beds is limited under such temperatures, which indicates that if ocean temperatures rise due to climate changes, the occurrence of rhodoliths could be limited in the tropical regions and restricted to higher latitudes, as suggested for other marine environments (Hooidonk et al., 2014; Wernberg et al., 2016; Vinagre et al., 2018). The likely vulnerability of tropical rhodoliths to extreme warming events and competition with epiphytes suggest that monitoring and conservation measures are fundamental in the face of environmental changes.

5. Conclusions

Temperatures, nutrients, current velocity and water transparency are among the main environmental drivers of rhodoliths and associated epiphytic assemblage structure. The abundance of rhodoliths followed the explored latitudinal gradient, with higher values in tropical sites decreasing towards warm temperate, corroborating to our first hypothesis. However, the suitable habitats indicated by the models were more abundant in the transition and warm temperate regions. The predictive model indicates a suitable area of around 230,000 km^2, with potential high biomass of rhodoliths 60 km off Paraná and São Paulo coast, a region that was not well explored yet.

The total epiphytes biomass have no clear latitudinal pattern, but the differences found in species composition corroborate with previous biogeographical subdivisions (Horta et al., 2001). Temperature and nitrate seems to be the major drivers controlling the composition of species, with warm-water adapted species in tropical region and cold-water and higher nutrient adapted species in warm temperate. Although in low abundance, tropical species not referred before for warm temperate beds appeared in summer 2015, which can indicate a tropicalization event. The diversity, abundance and ubiquitous presence of Rhodolith beds in the tropical and warm temperate coast of Southwestern Atlantic foster the implementation of further conservation initiatives of this organisms and environments, especially considering threats related with oil pollution, warming and ocean acidification.

Author contributions

VC, JS, JN, SB and PH planned the sampling design of the study. VC and JA performed the modelling and statistical analysis. PH, MB, AB, JS and JN were responsible for field collections and financial support. VC analyzed the field samples and wrote the first draft of the manuscript; JA, ES, JN, SB, JB, AB, MB and PH wrote sections of the manuscript. All authors contributed to manuscript discussion and revision. All authors approved the final article.

Declaration of competing interest

None.

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Appendix A. Supplementary data

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